

Fatty acids degradation and biosynthesis

1. Degradation of FAs in Mitochondria
2. Ketone body
3. Synthesis of FAs in Cytosol
4. Regulation of FAs metabolisms

Fatty acids

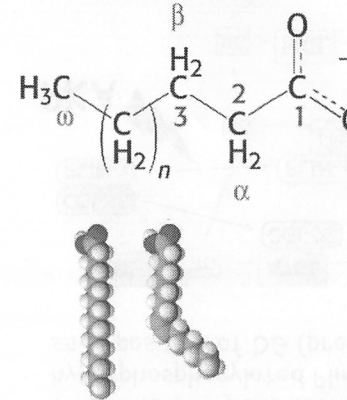


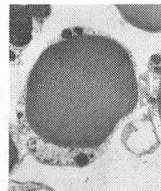
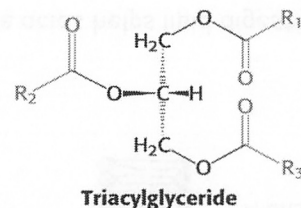
TABLE 17.1 • Fatty Acids of Importance to Humans

Name	Numerical Formula	Functions in Humans
Formic acid	1	
Acetic acid	2:0	
Propionic acid	3:0	Produced by metabolism of odd-chain fatty acids as well as isoleucine, valine, and methionine
Butyric acid	4:0	Milk triacylglycerols contain short chain fatty acids
Myristic acid	14:0	Covalently linked to some proteins
Palmitic acid	16:0	Product of fatty acid synthase
Palmitoleic acid	16:1(9)	Fatty acids with 16–18 carbons comprise the bulk of the fatty acids in triacylglycerols and complex lipids
Stearic acid	18:0	
Oleic acid	18:1(9)	
Linoleic acid	18:2(9, 12)	Essential fatty acid
Linolenic acid	18:3(9, 12, 15)	Essential fatty acid
Arachidonic acid	20:4(5, 8, 11, 14)	Precursor of prostaglandins and other eicosanoids
Lignoceric acid	24:0	
Nervonic acid	24:1(15)	Enriched in sphingolipids

From diet

1. Most FAs in human have 16–20 carbons in length.
2. They are fuel molecules stored as TG.
3. They are building blocks of glycolipids and phospholipids.
4. Many proteins are modified by covalent attachment of FAs, by which modified proteins are targeted to membrane fractions.
5. Their derivatives function as hormones and intracellular signaling molecules.

Triacylglycerol (TG)



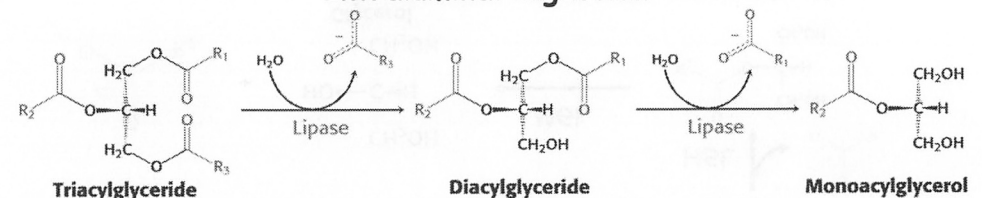
1. Complete oxidation of FAs yields about **9Kcal/g**, **twice larger** than that of proteins and carbohydrates.
2. It is non-polar and stored in nearly **anhydrous** form.
1g of glycogen binds about 2g of water.



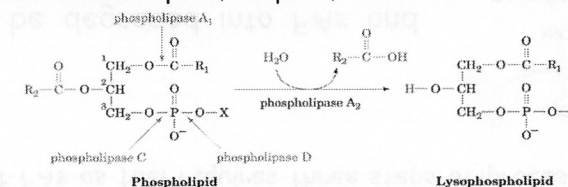
A gram of fat stores more than **6 times** as much energy as a **gram of glycogen**

3. It is stored in **cytosol** of adipocytes.
4. **Amounts of TG** in the body allow survival for **several weeks**, while those of **glycogen for about one day**.
5. It consists of **over 90%** of fat ingested (60–150g/day).

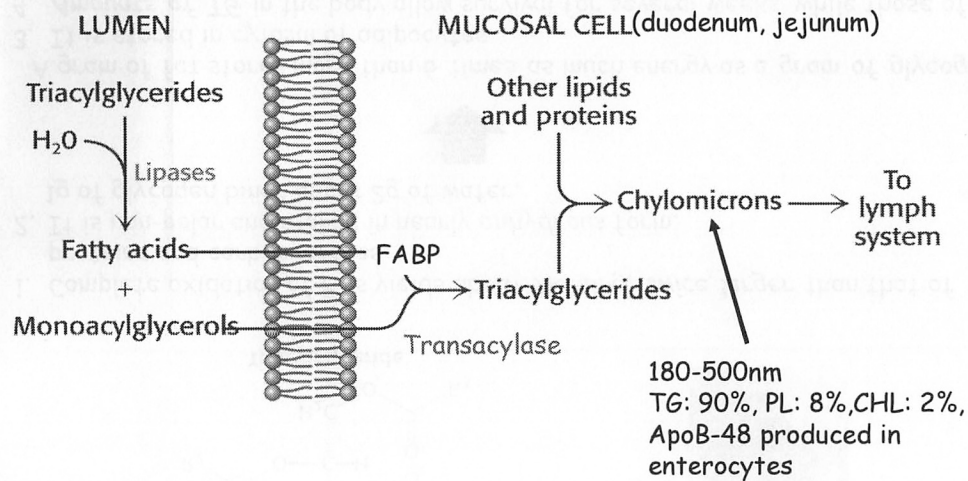
Intraluminal digestion



- First hydrolysis of TG (30%) occur in stomach by gastric lipase.
- **Pancreatic lipase is a major enzyme for TG hydrolysis**, which occurs in small intestine (duodenum, jejunum)
- **Bile acids**, that are synthesized from **cholesterol** in liver (**0.8g/day**), function as a detergent to make micelles.
- Colipase anchors lipase to TG in micelles, and protects lipase from bile acids-dependent denaturation.
- Pancreatic juice contains lipase, colipase, PLA2 and bile acids.

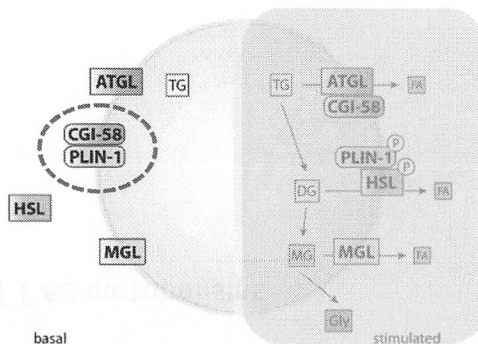
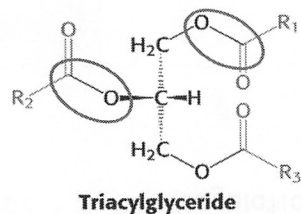


Intracellular processing of lipids and their secretion into lymph



Bile acids help lipid digestion as well as lipid absorption.

Intracellular TG hydrolysis



• **ATGL**, also known as a patatin-like phospholipase domain containing A2 (PNPLA2), act as a major **TG** hydrolase.

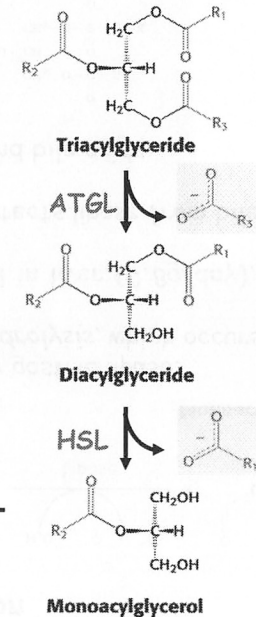
• **PNPLA** gene family has nine isoforms (PNPLA1~9), among which SNP of PNPLA3 is strongly associated with hepatic steatosis.

• **ATGL** preferentially cleaves fatty acyl ester at *sn-2* or *sn-1* position of **TG**.

• In the absence of PKA activation, perilipin1 (Plin1) appears to reduce lipolysis by binding and sequestering comparative gene identification-58 (**CGI-58**), a LD-associated protein, which can associate with and activate **ATGL**.

Utilization of FAs as fuel requires three steps of processing in the cell

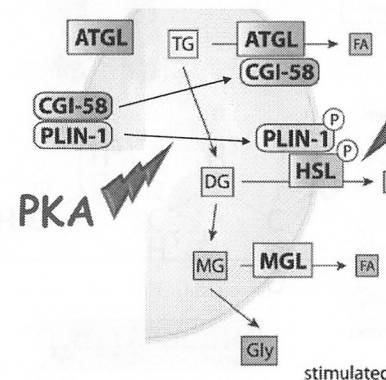
Lipids must be degraded into **FAs** and **glycerol** during fasting (**lipolysis**), which are released from **white adipose tissues** and transported to energy-requiring tissues such as heart, skeletal muscles, liver or brown adipose tissue.



Intracellular TG hydrolysis

• When **PKA** is activated by hormone such as epinephrine and glucagon, **PKA** hyperphosphorylates **Plin1**, resulting in the release **CGI-58**. Free **CGI-58** consequently binds to and activates **ATGL**.

• **PKA** activation also phosphorylates **HSL** (predominantly located in cytosol of WAT and BAT) and translocates cytosolic **HSL** to LD, where it binds hyperphosphorylated **Plin1** and hydrolyzes **DG** and cleaves fatty acyl ester at *sn-3* position of **DG** (production of **FA** and *sn-1* or *sn-2* **MG**).

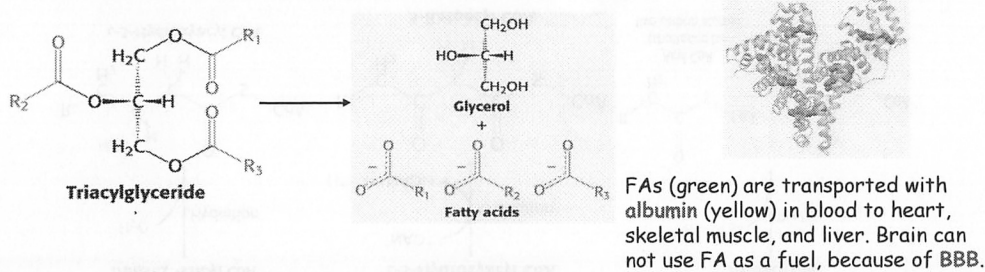


• **MG** lipase is ubiquitously expressed with highest levels in adipose tissues.

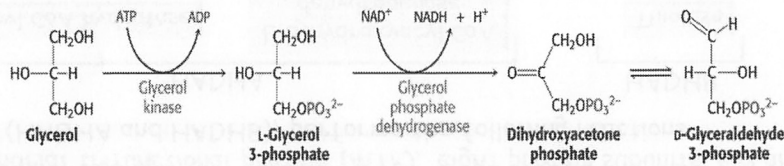
• Lipolysis in non adipose tissues is not well defined.

• In liver, **Plin2** but not **Plin1** is expressed and regulates **ATGL** activity.

FAs and glycerol are delivered to peripheral tissues

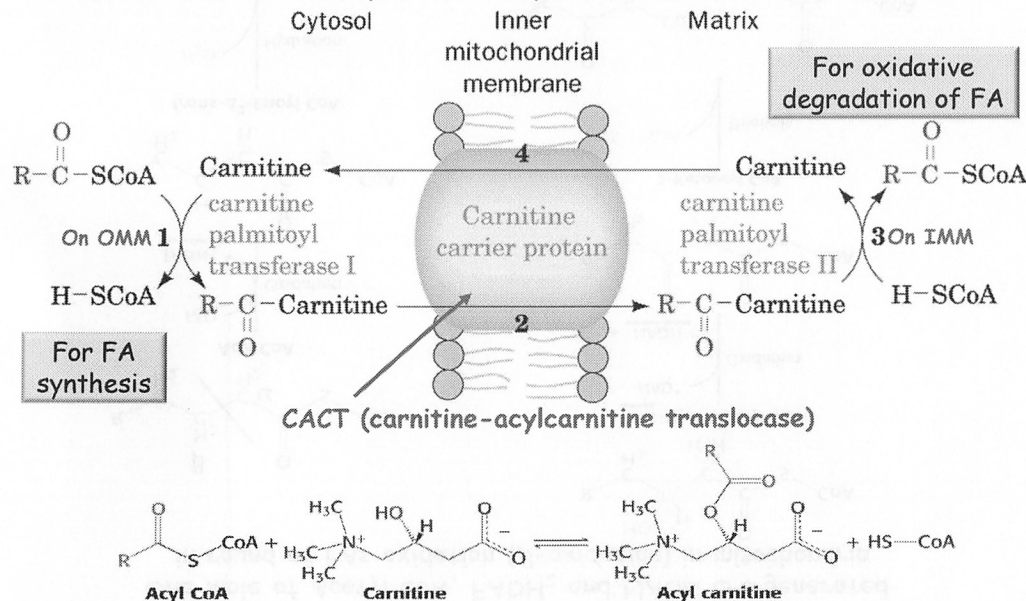


- *Glycerol is transported exclusively to the liver*, and is metabolized to pyruvate or glucose depending on energy status through DHAP.
- Liver has high activity of glycerol kinase.

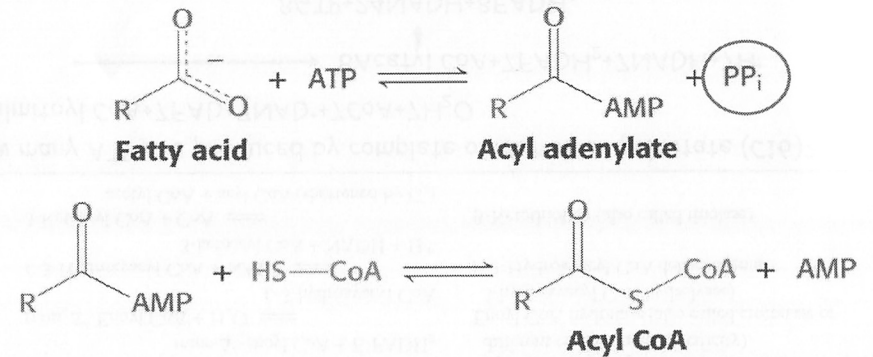


Carnitine carries long-chain (C12-C18)

but not medium-chain (passive transport) activated FAs into matrix



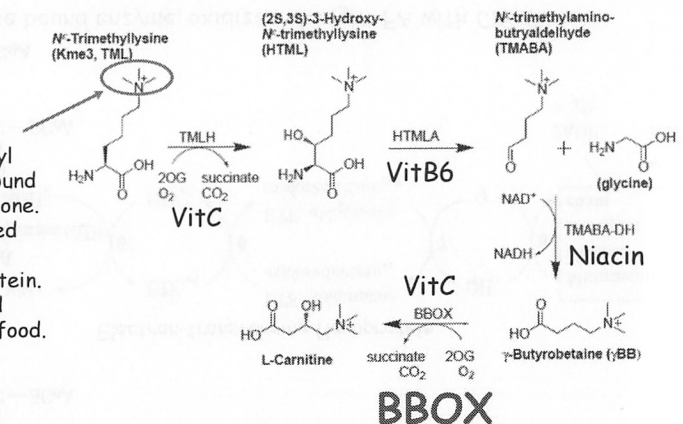
FAs are activated by *acyl CoA synthetase* before they enter mitochondria matrix



- Acyl CoA synthetase is present on outer mitochondria membrane (OMM)
- There are at least 26 kinds of ACS which have different affinities for distinct length of FAs, e.g. short, medium, long, and very-long chain FAs
- These above reactions are driven by hydrolysis of PP_i by pyrophosphatase

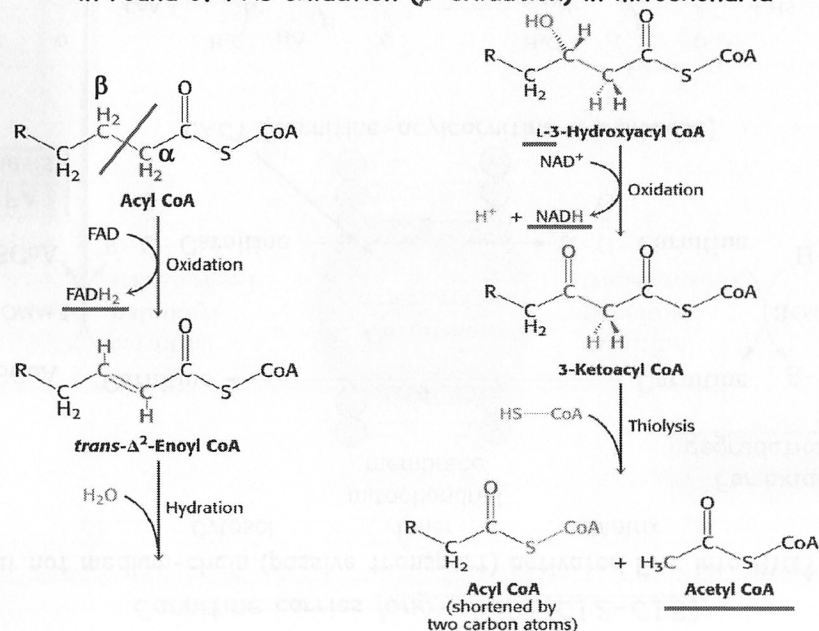
Carnitine Synthesis

Carnitine synthesis occurs predominantly in liver and kidney that express BBOX, from where carnitine is delivered to other tissues such as skeletal muscle and heart. Human can absorb carnitine from food (rich in red meat) .

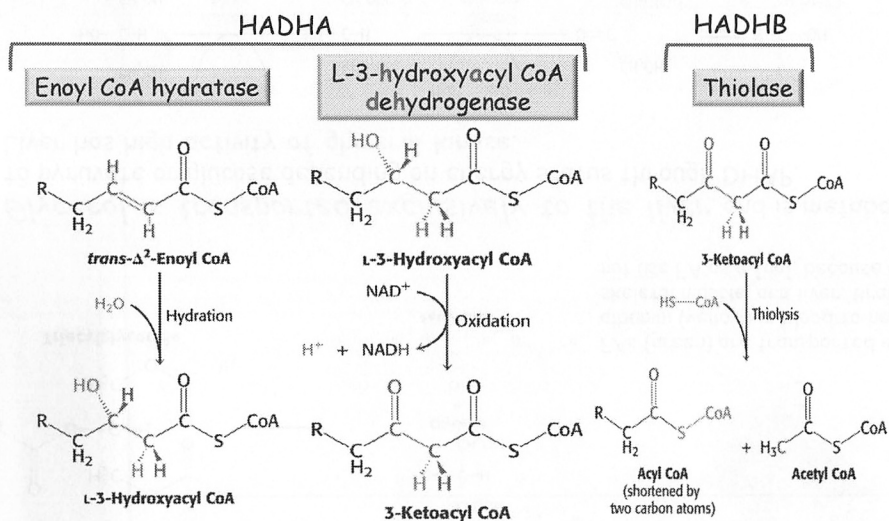


TML: N-trimethyllysine, TMLH: TML hydroxylase, HTMLA: hydroxyTML aldolase, BBOX: γ-butyribetaine hydroxylase

One mole of Acetyl CoA, FADH₂ and NADH are generated in round of FAs oxidation (β -oxidation) in mitochondria

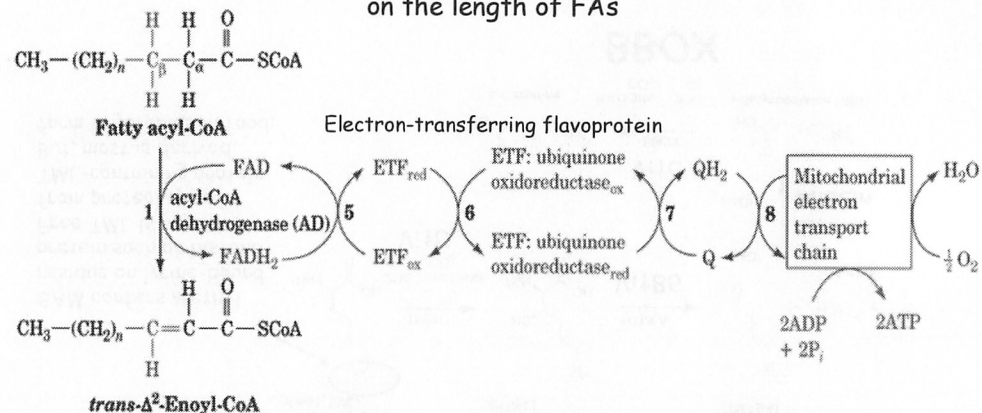


Mitochondrial trifunctional protein (MTP), eight protein subunits, 4 α and 4 β subunits (HADHA and HADHB), performs the following reactions



The enzyme for long-chain FA is associated with mitochondrial inner membrane, while the enzyme for medium and short FA is present in matrix.

Acyl CoA dehydrogenase has at least four different types depending on the length of FAs

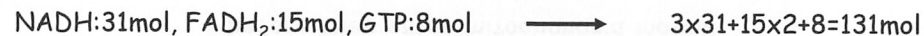
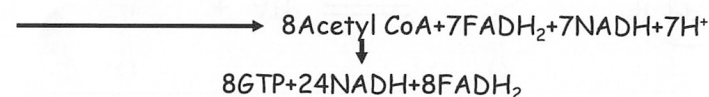
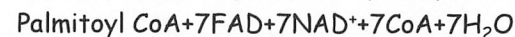


- VLCAD, a membrane bound enzyme, oxidizes straight FA with C12-C24.
- LCAD is involved in branched chain FA oxidation.
- MCAD shows broad chain-length specificity with most active for C6 and C8 substrate.
- SCAD shows higher activity for substrate with C4>C6>C8.

TABLE 22.1 Principal reactions in fatty acid oxidation

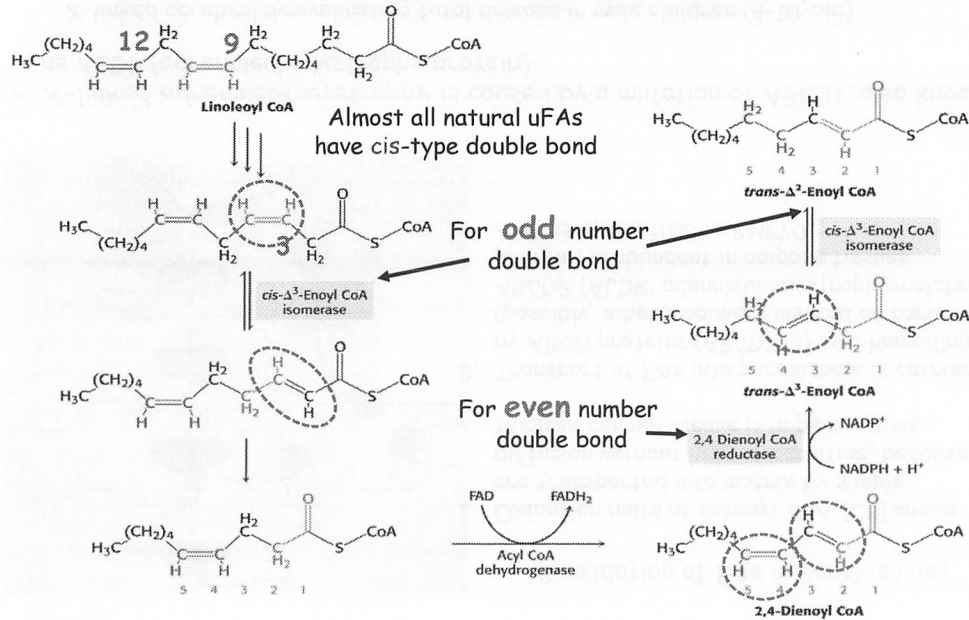
Step	Reaction	Enzyme
1	Fatty acid + CoA + ATP $\rightleftharpoons$ acyl CoA + AMP + PP _i	Acyl CoA synthetase [also called fatty acid thiokinase and fatty acid:CoA ligase (AMP)]
2	Carnitine + acyl CoA $\rightleftharpoons$ acyl carnitine + CoA	Carnitine acyltransferase (also called carnitine palmitoyl transferase)
3	Acyl CoA + E-FAD $\rightarrow$ trans- Δ^2 -enoyl CoA + E-FADH ₂	Acyl CoA dehydrogenases (several isozymes having different chain-length specificity)
4	trans- Δ^2 -Enoyl CoA + H ₂ O $\rightleftharpoons$ L-3-hydroxyacyl CoA	Enoyl CoA hydratase (also called crotonase or 3-hydroxyacyl CoA hydrolyase)
5	L-3-Hydroxyacyl CoA + NAD ⁺ $\rightleftharpoons$ 3-ketoacyl CoA + NADH + H ⁺	L-3-Hydroxyacyl CoA dehydrogenase
6	3-Ketoacyl CoA + CoA $\rightleftharpoons$ acetyl CoA + acyl CoA (shortened by C ₂)	β -Ketothiolase (also called thiolase)

How many ATP are produced by complete oxidation of palmitate (C16)?



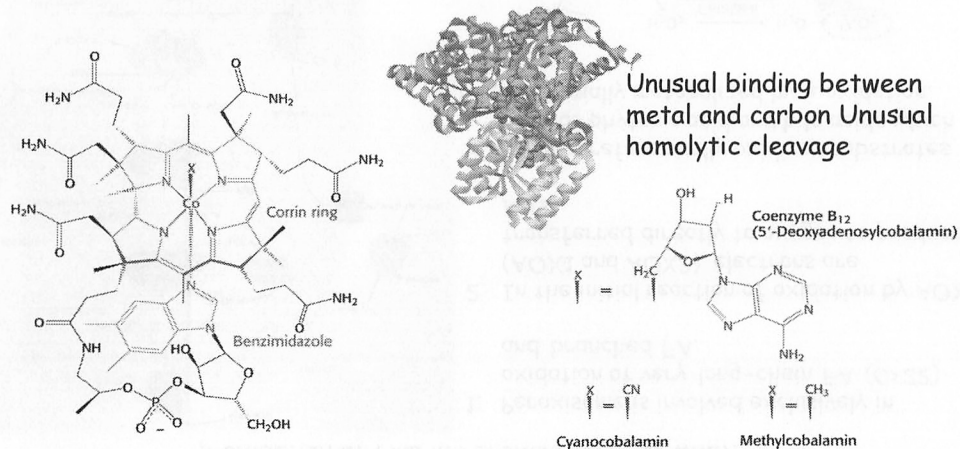
131-2=129mol ATP/palmitate oxidation

unsaturated FAs (uFAs) oxidation

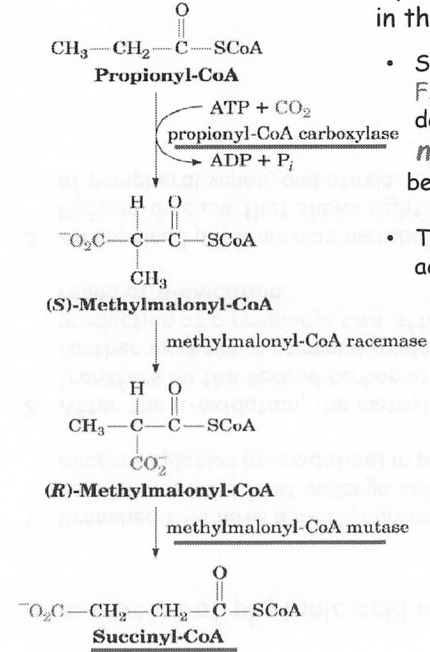


Cobalamin (vitamin B₁₂ derivative) containing enzyme catalyzes three distinct type of reactions

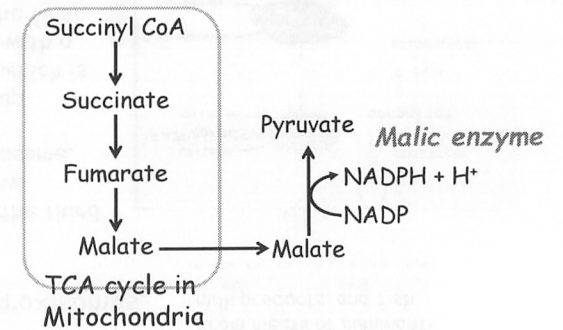
1. Intramolecular rearrangement as seen in conversion of L-methylmalonyl CoA into succinyl CoA. (5'-deoxyadenosylcobalamin)
2. Methionine synthesis from homocysteine. (methylcobalamin)
3. Reduction of ribonucleotides to deoxyribonucleotides (**not in mammals**).



Odd number chain FAs produce **propionyl CoA**, but not acetyl CoA in the final thiolysis step.

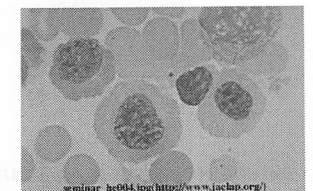


- Succinyl CoA produced in the course of odd-number FA oxidation or amino acids (Val, Ile, Met, Thr) degradation from methylmalonyl CoA is *not oxidized directly in TCA cycle*, because succinyl CoA serves as a catalyst in TCA cycle.
- To oxidize succinyl CoA, it must be metabolized to acetyl CoA via pyruvate.

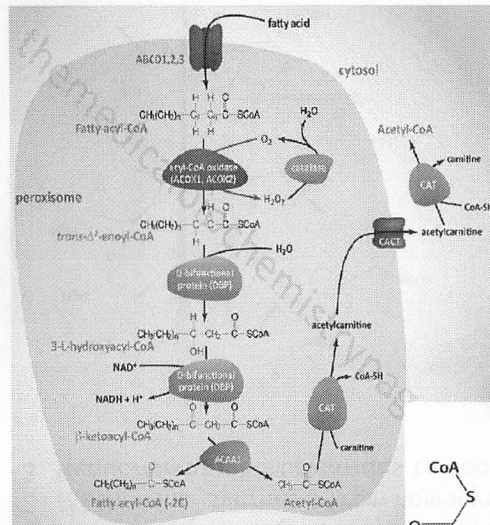


Deficiency in Vitamin B₁₂ causes severe anemia, known as *pernicious anemia* (惡性貧血).

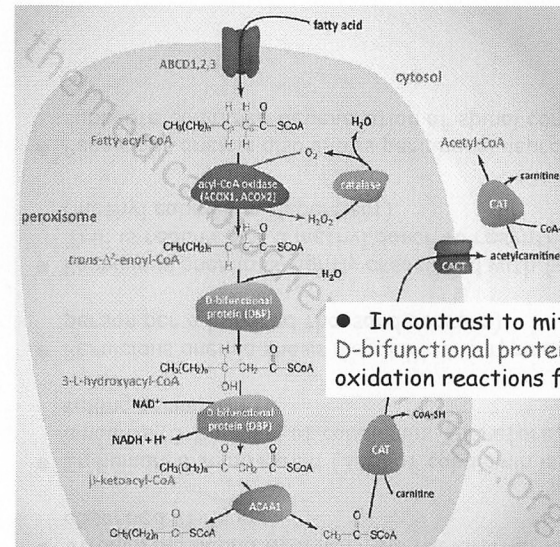
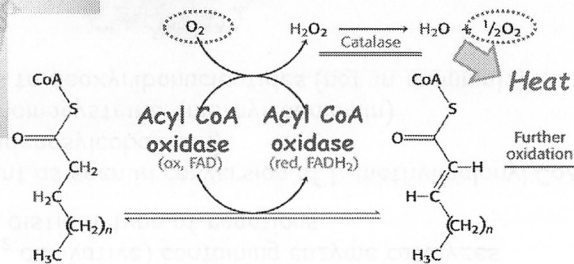
- Vitamin B₁₂ bound with intrinsic factor that is released from gastric parietal cells is absorbed at ileum.
 - In human, a small amount (3μg) of cobalamin is daily required and liver stores larger amounts (3-5 years) of cobalamin. Therefore, a lack of vitamin B₁₂ seldom occurs when taking normal diet.
 - Pernicious anemia shows megaloblastic type anemia is prevalent in elder women (1~5 person per a hundred thousand persons).
 - Pernicious anemia is tightly associated with folate deficiency, because N⁵-methyl THF is required as a methyl donor to resynthesize methionine by methionine synthase (methyl cobalamin-dependent).
 - Pernicious anemia also shows peripheral neuropathy and subacute combined degeneration of spinal cord.
- 



β-oxidation of FAs in Peroxisomes. For what?



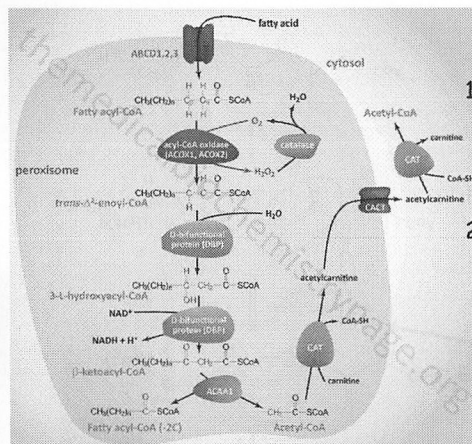
1. Peroxisome is involved exclusively in oxidation of **very long-chain FA (C>22)** and **branched FA**.
2. In the initial reaction of oxidation by AOX (AOX1 and AOX2), electrons are transferred directly to oxygen to produce H_2O_2 .
3. AOX2 preferentially oxidizes substrates such as phytanic acid and bile acids which are initially metabolized by α oxidation.



β-oxidation of FAs in Peroxisomes

- In contrast to mitochondrial FA oxidation, D-bifunctional protein (DBP) carries out both hydration and oxidation reactions for substrates except dicarboxylic acid.

- In the final step, acetyl CoA C-acyltransferase (ACAA1, peroxisomal thiolase) and sterol carrier protein x (SCPx) cleave and shorten in C2. Especially, SCPx is involved in thiolysis of α-oxidized branched FAs and in Bile acid formation.



β-oxidation of FAs in Peroxisomes

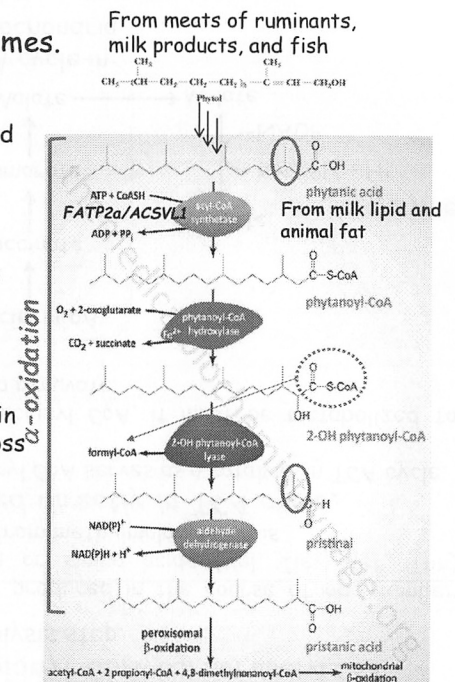
1. Oxidation halts at octanoyl CoA (C8) which are transported into matrix by simple diffusion without help of carnitine, because thiolase can not cleave it in peroxisome.
2. Transport of FAs into peroxisome is carried by ABCD proteins (ABCD1~3) as a homodimer (possibly a heterodimer) instead of carnitine. ABCD2 (ALDR: adrenoleukodystrophy-related protein) is abundant in adipose tissues. ABCD3 is identical to PMP70.

- **X-linked adrenoleukodystrophy** is caused by a mutation of ABCD1, also known as ALDP (adrenoleukodystrophy protein)

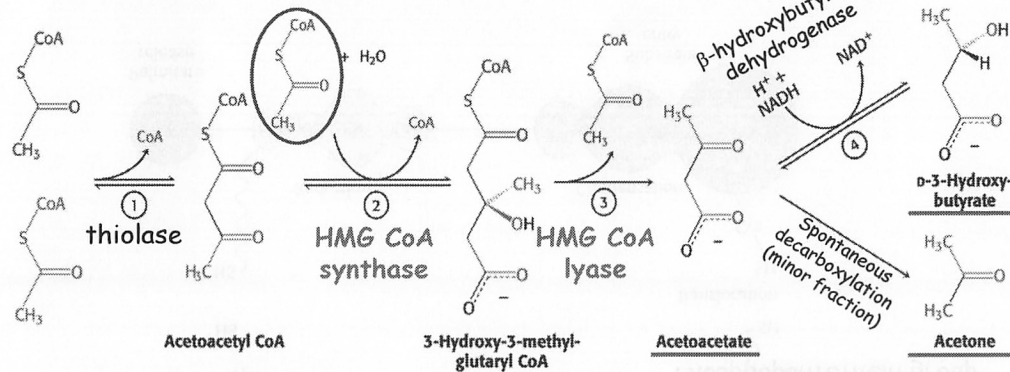
X-linked cerebral demyelinating fatal disease in male children (4-10yold). Behavioral abnormality, dementia, speech difficulty, loss of vision and hearing. Accumulation of very long FAs in blood and destruction of myelin sheath.

α-oxidation of phytanic acid in Peroxisomes.

1. Branched FAs have a **methyl** group on the third carbon atom and first undergo **oxidative decarboxylation** (α-oxidation) in peroxisome.
2. After the α-oxidation, the methyl group transfers on the second carbon atom, which is further oxidizes in normal β-oxidation with a production of a **propionyl CoA** after the first round of β-oxidation.
3. An impaired phytanic acid metabolism results in **Refsum disease** that shows night blindness, loss of peripheral vision, and ataxia.

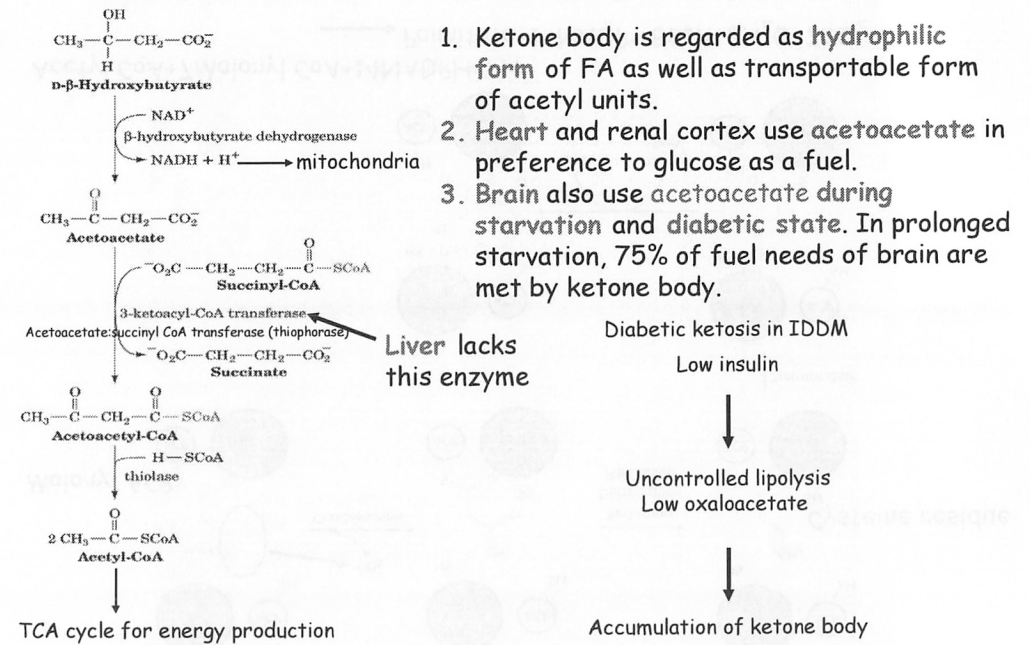


Ketone body, an important metabolic fuel, is formed by Acetyl CoA.

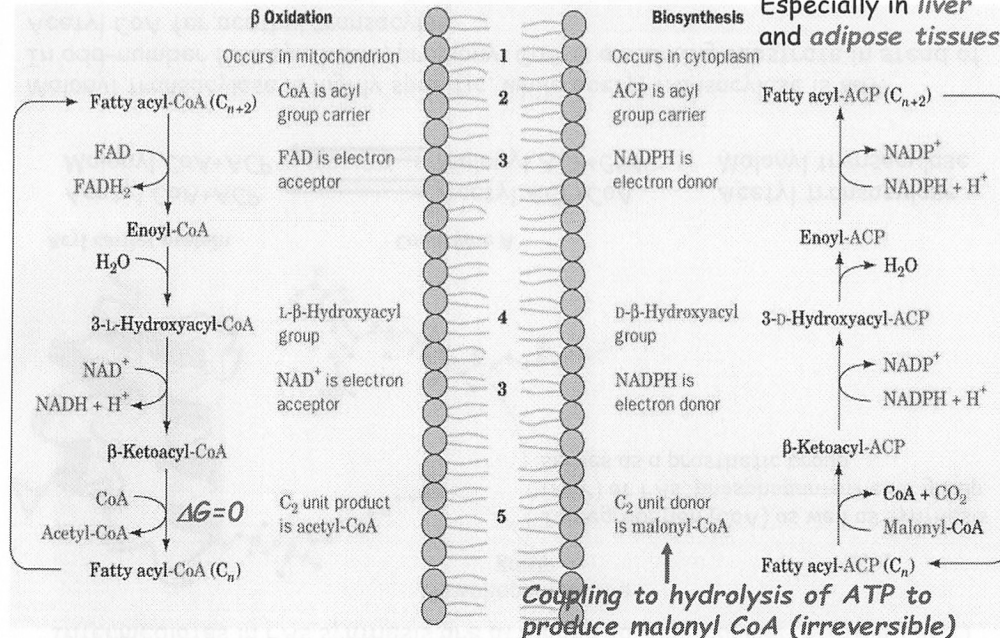


- β-hydroxybutyrate dehydrogenase reaction is reversible depending on ratio of NAD⁺/NADH in mitochondria. This enzyme is highly expressed in **liver**. Under fasting conditions, equilibrium is toward to formation of 3-OHB.
- Ketone body is generated in **liver (major)** and kidney cortex mitochondria matrix **during prolonged fasting**.
 - Mitochondria in only hepatocytes, but not other cells, have **HMG CoA synthase**.
 - HMG CoA synthase is also present in **cytosol** of hepatocytes for **cholesterol synthesis**.
 - HMG CoA lyase** is expressed only in hepatic (and kidney) mitochondria.

Acetoacetate is transformed into 2mol Acetyl CoA in peripheral tissues

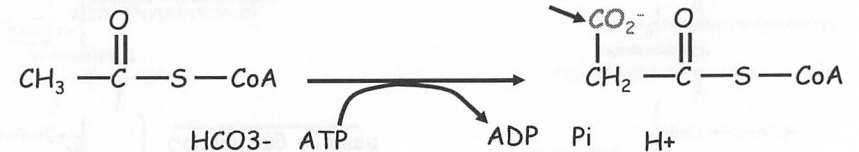


What is main difference?



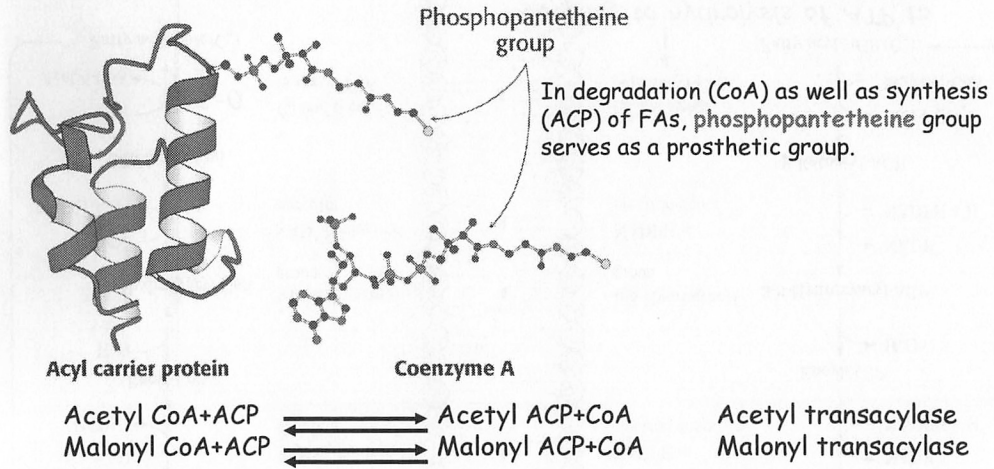
Formation of malonyl CoA is first and rate-limiting step in FAs synthesis catalyzed by Acetyl CoA carboxylase (ACC)

This carbon dose not appear in the synthesized FAs product



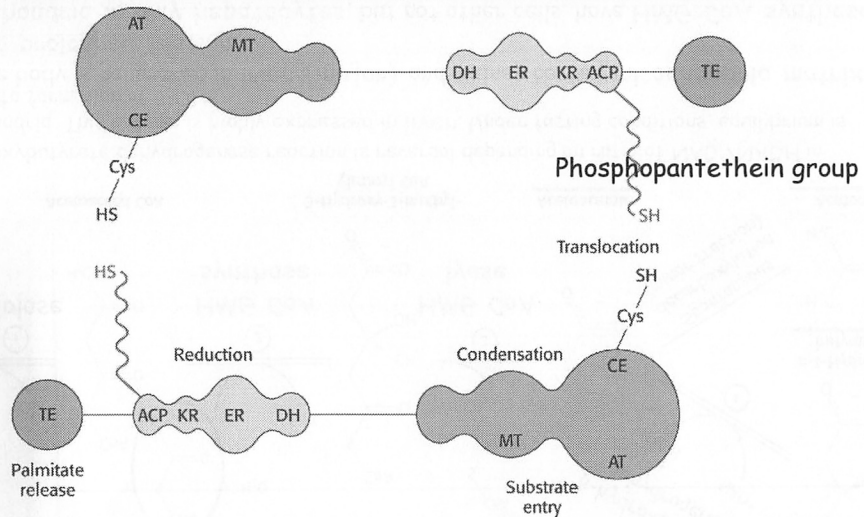
- It is a biotin containing enzyme.
- It has two major isoforms, α and β-ACC.
 - α-ACC (ACC1, cytosolic) is involved in FAs synthesis and is expressed exclusively in liver and adipose tissue.
 - On the other hand, β-ACC (ACC2, mitochondrial) expressed only in tissues that oxidize but not synthesize FAs, such as heart and skeletal muscles. It functions as a regulator for FAs oxidation via inhibitory action of malonyl CoA on transport of FAs into mitochondria (CPT-I), but not for FAs synthesis.
- It is subject to hormonal regulation (PKA) and energy regulation (AMPK) as well as its polymerizational regulation.

Intermediates in FAs synthesis are attached to Acyl carrier protein (ACP)



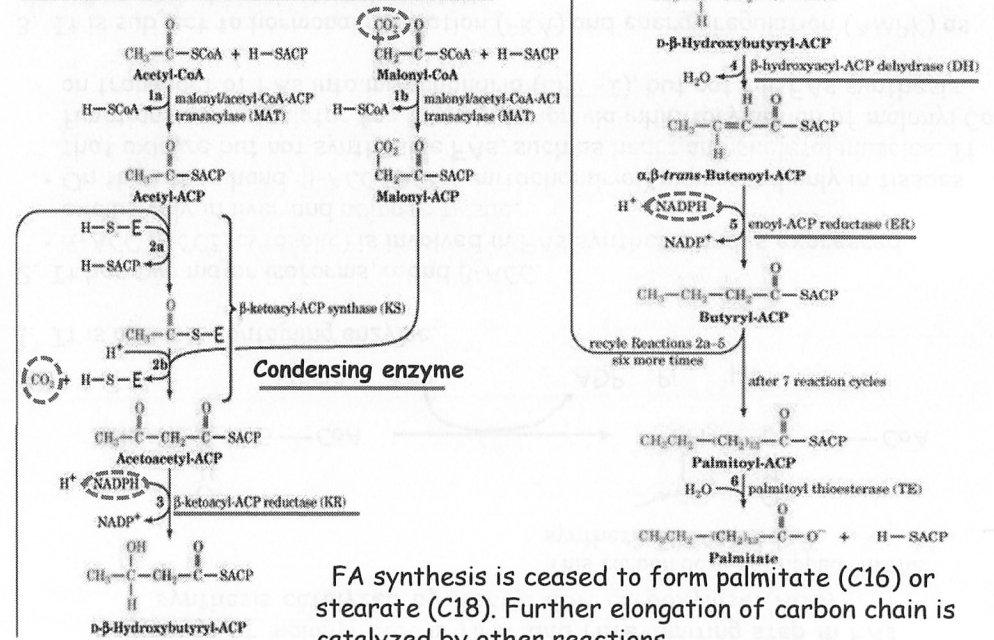
Malonyl transacylase is highly specific, while acetyl transacylase is not. In odd-number FAs synthesis, propionyl CoA is a starting substrate in stead of Acetyl CoA for acethyl transacylase.

Mammalian FA synthase (FASN), a multifunctional enzyme complex, has seven catalytic domains and functions as **homodimer**

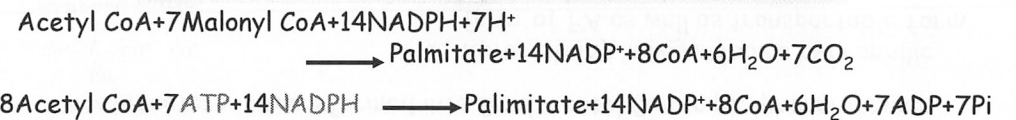
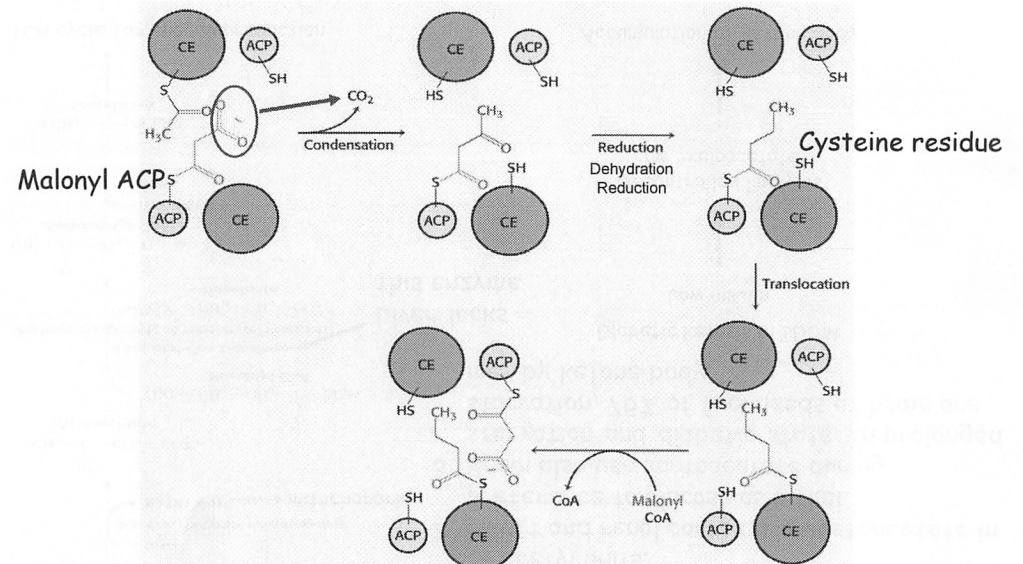


AT: acetyl transacylase MT: malonyl transacylase CE: condensing enzyme KR: β -ketoacyl-ACP reductase
DH: β -hydroxyacyl-ACP dehydrase ER: enoyl-ACP reductase TE: palmitoyl thioesterase

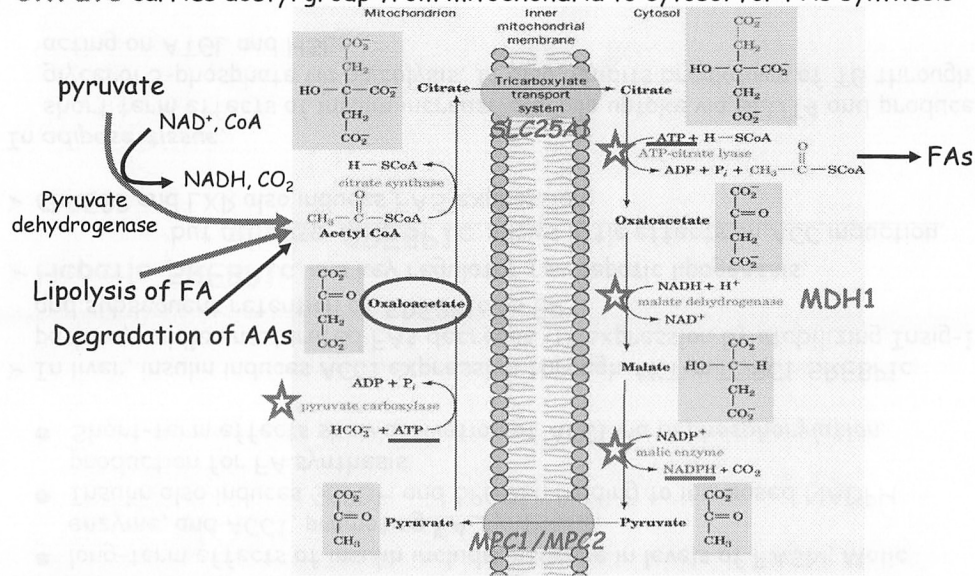
Principle reactions in FAs synthesis



FA synthesis is ceased to form palmitate (C16) or stearate (C18). Further elongation of carbon chain is catalyzed by other reactions



Citrate carries acetyl group from mitochondria to cytosol for FAs synthesis

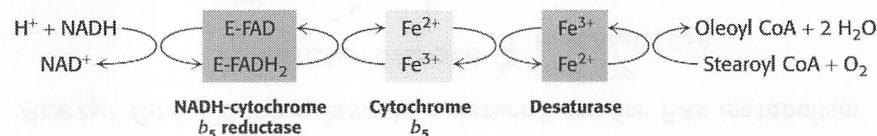
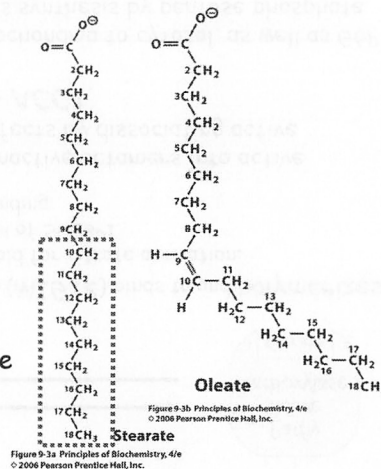


- 1mol of NADPH is produced when 1mol Acetyl CoA is transported to cytosol.
- In synthesis of palmitate, 8mol NADPH are formed by this process. Additional 6mol NADPH required for the synthesis come from the pentose phosphate shunt.

Unsaturation of FAs

- Introduction of unconjugated double bond is catalyzed by a complex of three ER enzymes.
- Double bond can not be inserted during FA synthesis until C16 or C18 is formed.
- Mammalian enzymes can not introduce double bond at carbon atom *beyond C9*.

Stearoyl-CoA desaturase is an important enzyme



Elongation of FAs

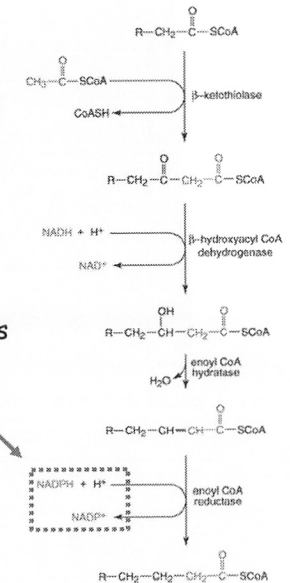
- Elongation of FAs is achieved by adding two carbon units from Malonyl CoA, but not from Malonyl ACP on ER.

Preferred substrate is palmitoyl CoA.

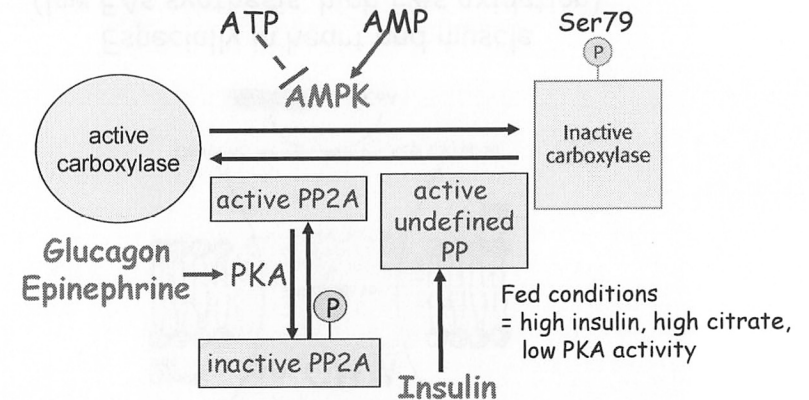
- It also occurs in mitochondria by reversal reactions of FAs oxidation with an exception that NADPH takes the place of FADH₂ as a terminal redox coenzyme in the enoyl CoA reductase reaction.

In this case, acetyl CoA is a donor for two carbon.

Preferred substrate is shorter fatty acyl CoA (<C16)

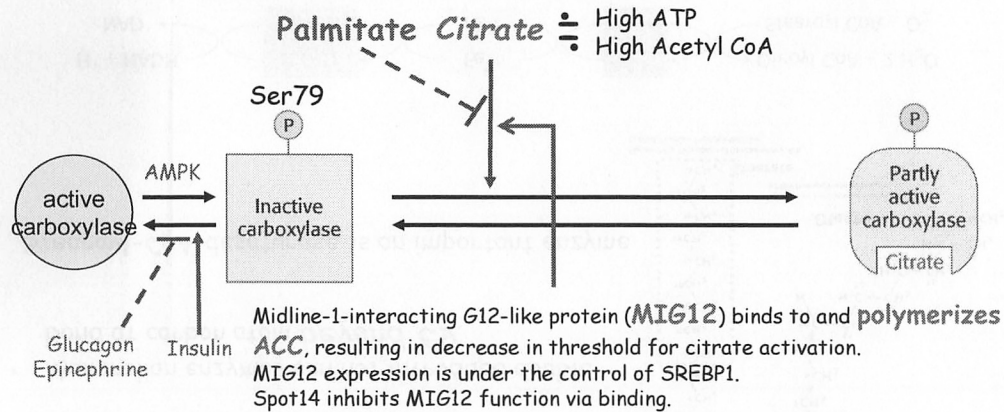


Acetyl CoA Carboxylase is a determinant for FAs metabolism



- α-ACC (ACC1, cytosolic) is involved in FAs synthesis and is expressed exclusively in liver and adipose tissue.
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Acetyl CoA Carboxylase is a determinant for FAs metabolism



- Citrate (supraphysiological levels > 5mM) polymerize inactive octamers into active filaments, while palmitate inhibits this stimulatory effects by dissociating active filaments. These regulations predominantly occur for **ACC1**.
- Palmitate also inhibits transport of citrate from mitochondria to cytosol, as well as G6P dehydrogenase activity that produces NADPH for FAs synthesis by pentose phosphate shunt.

Effects of insulin

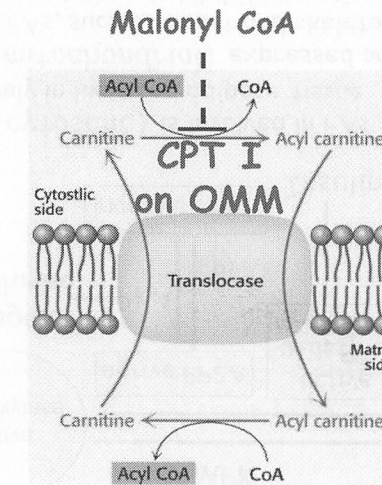
In liver,

- long-term effects of insulin include increase in levels of FASN, Malic enzyme, and ACC1, promoting **FA synthesis**.
 - Insulin also induces G6PDH and 6PGDH, leading to increased NADPH production for FA synthesis.
 - Short-term effects show activation of ACC1 via **dephosphorylation**.
- In liver, insulin induces ACC1 expression through AKT-mTORC1-SREBP1c pathway, while unsaturated FAs decrease its expression by stabilizing Insig-1 and subsequent retention of SREBP1c on ER.
- **Hepatic SREBP1c** is a key regulator for hepatic lipogenesis, but **adipose SREBP1c** shows little effects on ACC induction.
- ChREBP and LXR also induces FAS expression.

In adipose tissue,

short term effects of insulin increase glucose uptake via GLUT4 and produce glycerol 3-phosphate via glycolysis. It also inhibits breakdown of TG through acting on ATGL and HSL.

Malonyl CoA, produced by ACC2, inhibits FA oxidation



Especially in heart and muscle
(low FAs synthesis, high FAs oxidation)

ω -oxidation of phytanic acid in Peroxisomes.